UCI/INRF IGZO TFT PROCESS

A PDK for the IGZO TFT process

What was built, what it can do, and how far the numbers can be trusted.

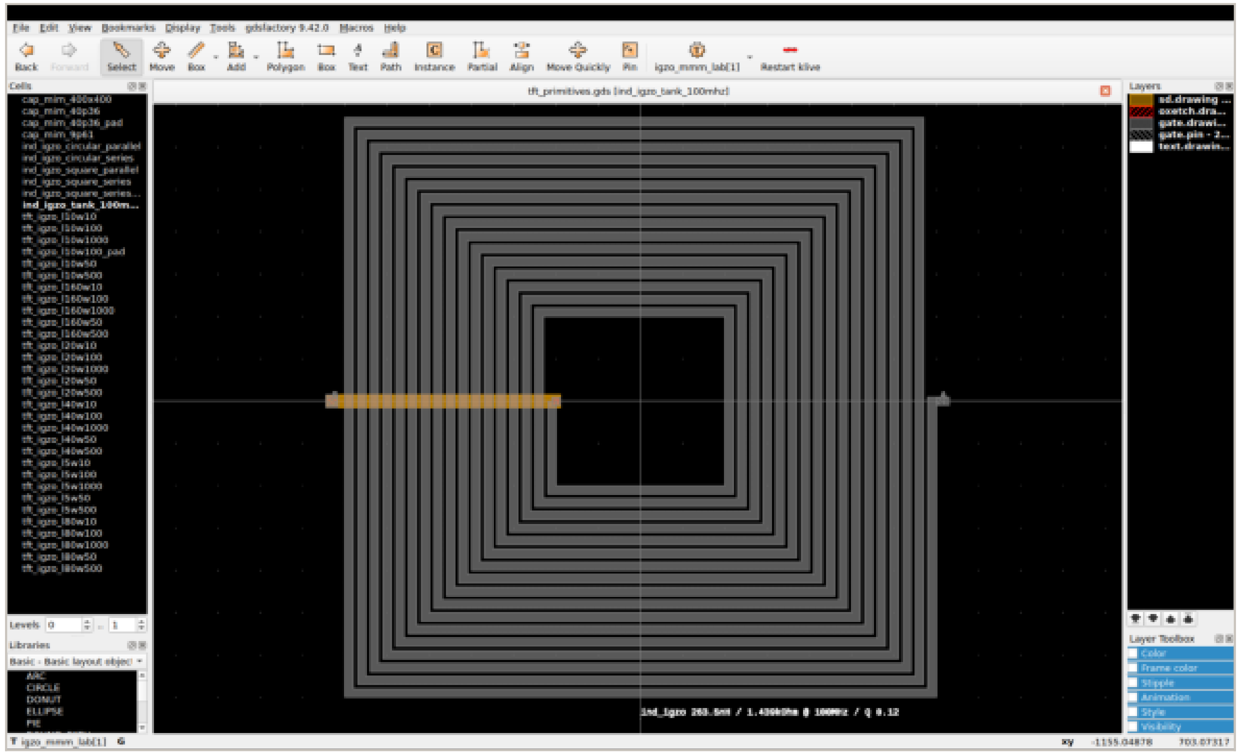
github.com/Juander28/TFT-MMM-LAB-PDK

- Four masks, top-gate coplanar, measured on three fabricated chips
- Used like sky130A or gf180mcuD: one command switches every tool
- Three parametric cells: transistor, capacitor, inductor
- Fifteen automated checks across klayout, xschem, ngspice, magic, netgen

One command, four tools

use-pdk TFT-MMM-LAB-PDK

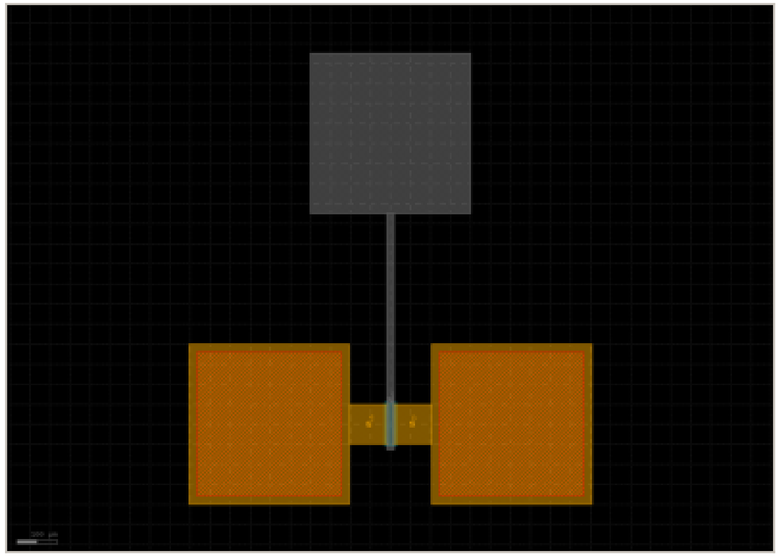
- klayout - technology, layer colours, the PCell library, DRC
- xschem - symbols and testbenches
- ngspice - the models and their corners
- magic - layout, extraction and the .mag device views
- netgen - LVS between the two



The cell library open in KLayout: 41 generated cells, the technology selected, the process's own layer names on the right

The transistor, and what each parameter does

tft_igzo		
w_gate	100 um	width of one finger; the total is w_gate x nf
l_gate	10 um	the DRAWN channel length - the gap between electrodes
nf	4	fingers. They share electrodes: nf fingers, nf+1 electrodes
gate_ov	5 um	gate overlap per side - the parasitic that decides fT
sd_len	97.5 um	
s_strap_w	40 um	width of the strap tying the source fingers together
d_strap_w	20 um	and the drain side, separately - they carry different current
pad	false	
l_bias	3 um	
l_printed	13 um	read-only: what the process actually prints, l_gate + 3 um
w_total	400 um	

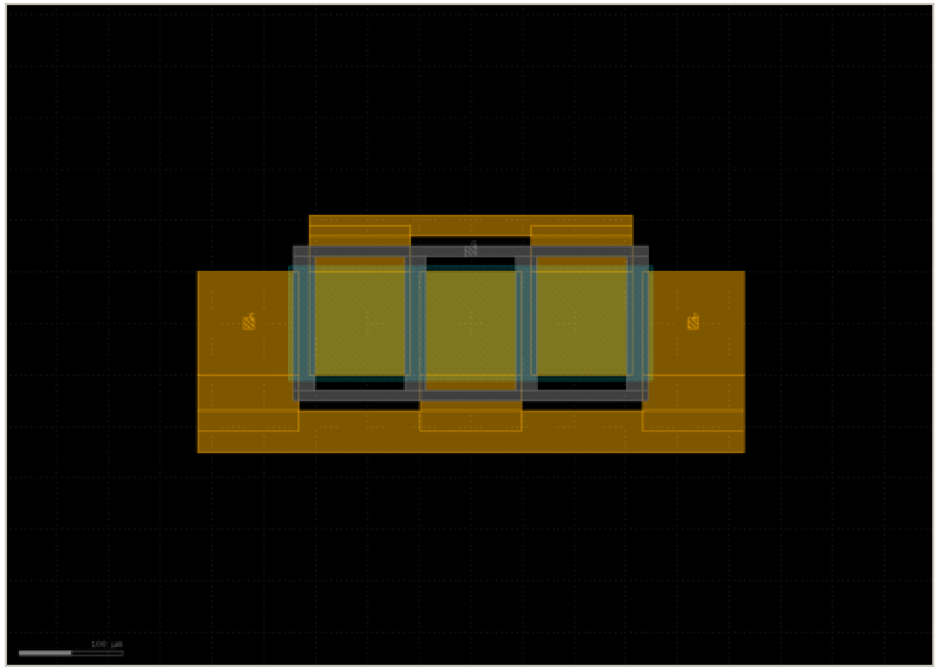


tft_igzo, four fingers, drawn by the PCell.

Fingers are only a layout choice

Splitting the width folds the device and divides gm and Cgs by the same number, so their ratio - and fT - do not move. Measured, then confirmed in simulation for nf = 1 to 20 at constant total width.

Fingers buy aspect ratio and gate resistance. Not speed.

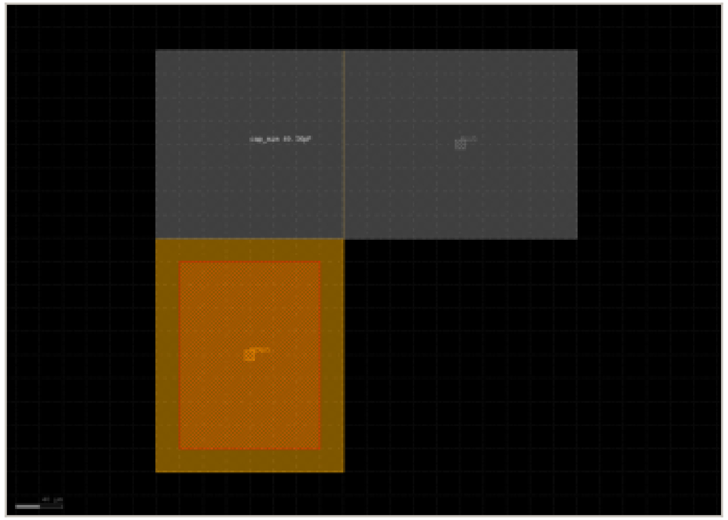


Four fingers with a 40 um source strap and a 20 um drain strap. Unstrapped this is five separate terminals, not two.

The capacitor sizes itself

One formula, C = Cox x W x L, solved in whichever direction the problem arrives in. Cox = 1.564 fF/um^2 is measured, not assumed.

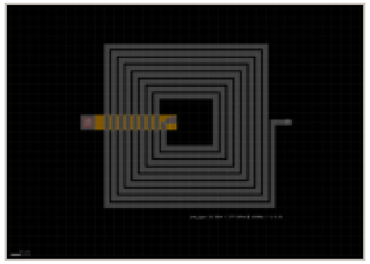
cap_mim		
mode	by capacitance	set the plate, the area, or the capacitance you need
w	160.64 um	...and the cell solves the plate: 160.64 um square
l	160.64 um	
area_target	25805 um^2	
c_target	40362 fF	ask for 40362 fF...
ext_sd	200 um	how far the S/D plate runs to its terminal
ext_gate	200 um	the gate plate has its own number - different metal, different route
ext_sd_far	0 um	and each can run out the far side as well
ext_gate_far	0 um	
pad	true	
area	25805 um^2	
c	40359 fF	read-only, and what the cell prints on itself

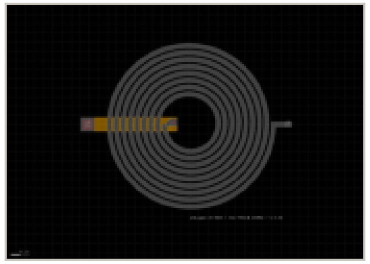


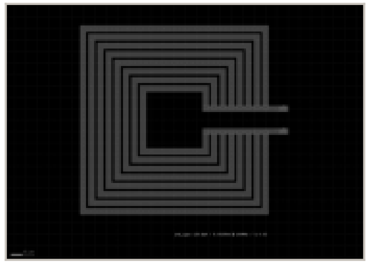
Sized by capacitance, with probe pads. Magic extracts 40.3593 pF from it - the drawn value

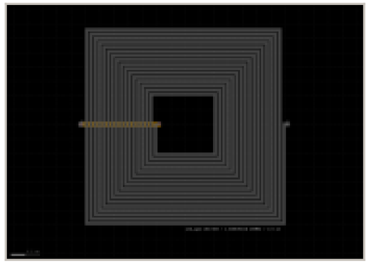
The inductor: shape, topology, and honesty

ind_igzo		
shape	square	square or circular
topology	series	one spiral in series, or n rings in parallel
n	8	turns, or rings
d_in	100 um	
w	10 um	
gap	5 um	
metal	gate (2/0)	
t_metal	0.05 um	metal thickness - ASSUMED, and the only real lever on Q
f_eval	100 MHz	
pad	false	
l_series	16.38 nH	read-only: Mohan-Wheeler, or coupled rings via Neumann
r_ac	377.7 Ohm	
q	0.03	read-only: 0.03 at 100 MHz, and the cell says so on its face









Q is a metal thickness problem

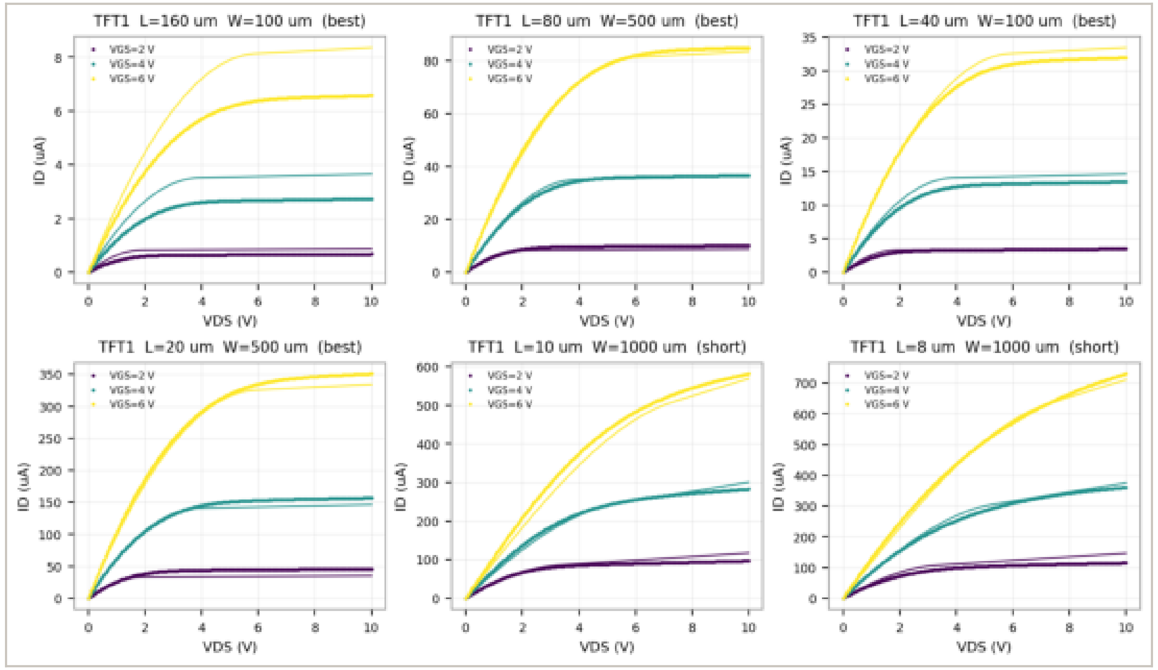
The same coil, the same 16.38 nH, thickness swept:

gold	R at 100 MHz	Q
50 nm (assumed today)	377.7 Ohm	0.03
250 nm	75.5 Ohm	0.14
1 um	18.9 Ohm	0.54

At 50 nm a planar inductor here is a resistor with some inductance.
Nothing you can draw changes that - only thicker gold does.

Simulated against measured

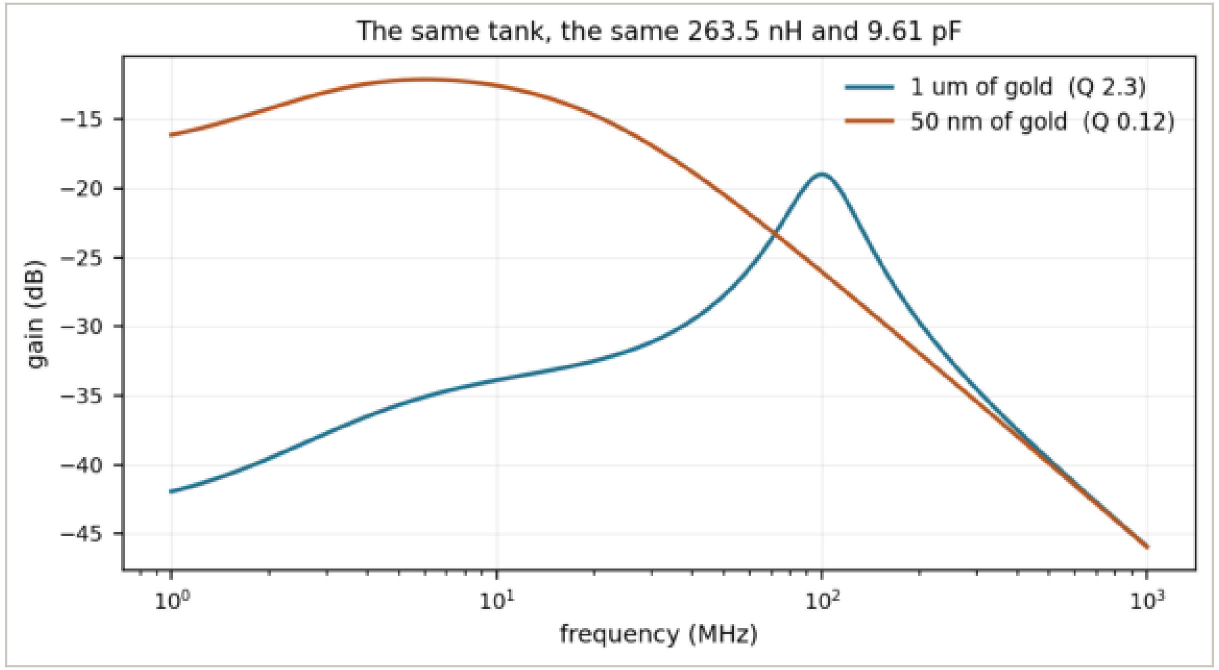
Points are MEASURED on the fabricated chips. Lines are SIMULATED in ngspice with the PDK's own corner models, at each device's own W and L. Six devices, L = 8 to 160 um.



Median simulated/measured in saturation: 1.01 over 18 comparisons.

The three cells together

TFT + cap_mim + ind_igzo as a tuned load, simulated in ngspice.
263.5 nH with 9.61 pF resonates at 100.0 MHz - if the gold is thick.



What holds this together

- The PCell reproduces the fabricated device shape for shape
- DRC clean on every generated cell but one, which is documented
- magic reads and writes the four-mask GDS, layer for layer
- LVS matches uniquely, three terminals on both sides
- Extracted capacitance is the drawn capacitance: 40.3593 pF of 40.36
- The model still reproduces the measurement: 654 uA against 670-701

Fifteen checks, one command: ./scripts/run_checks.sh

What is measured, and what is not

MEASURED	Cox = 1.564 fF/um^2, Kp, Vth, lambda, contact resistance
OFF THE CHIP	overlap 5 um, electrode sizes, pad and window sizes
ASSUMED	metal thickness 50 nm, the +3 um print bias, every Q
NOT KNOWN AT ALL	gate capacitance vs bias, fT, drift under stress

Every parameter in the SOP carries this label. A PDK reads as authoritative whether or not its numbers deserve it; the only defence is saying which is which, on every number.